
Does the Source of Carrier Image Affect Steganographic Detectability?

Final Presentation

Abdul Moiz Akbar | Daria Gjonbalaj | Nico Muller-Spath
Jimena Narvaez del Cid | David Wicker | Nikolas Zouros

Department of Advanced Computing Sciences
Maastricht University

Project 2.2 | May 2026

Agenda

1. Motivation recap
2. Research questions
3. Experimental design and methods
4. Statistical analysis
5. Results across RQ1–RQ5
6. Limitations and contribution

Motivation Recap

- Steganalysis detectability depends on carrier-image statistics, not only on embedding logic (Petitcolas et al., 1999; Fridrich & Kodovsky, 2012).
- Photographic carriers dominate benchmarks, yet diffusion outputs are now everywhere and carry distinct statistical traces (Corvi et al., 2023).
- Closest prior work studies AI carriers under adaptive embedding and reports P_E , not AUC (Méreux et al., 2024).

Core question

Do detectors validated on photographs remain equally effective when the carrier is ML-generated, under identical embedding settings?

Research Questions

RQ1 (Primary, confirmatory)

Does carrier source (real vs. pooled ML) affect detectability under matched embedding settings?

RQ2 (Primary, confirmatory)

Within ML carriers, does the choice of generator (SDXL vs. FLUX) change detectability?

RQ3 (Exploratory)

Does payload size change the detectability gap between real and ML carriers?

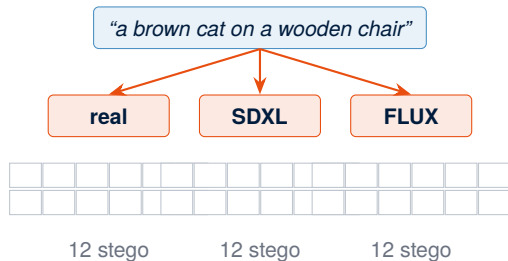
RQ4 (Exploratory)

Do the spatial (LSB+PNG) and frequency (DCT-LSB+JPEG) branches show different carrier-source effects?

RQ5 (Verification)

Does AES-256-CBC payload encryption affect detectability? Expected null result.

Caption-Matched Carriers, $3 \times 2 \times 3 \times 2$ Factorial



Per-group recipe

- Three caption-matched covers per group
- $2 \text{ methods} \times 3 \text{ payloads} \times 2 \text{ encryption} = \mathbf{12 \text{ stego per source}}$
- Per group: **3 cover + 36 stego** = 39 images

Datasets (3,000 groups; 1,000 pre-registered)

- Real: COCO + Flickr30k captions
- ML-A: SDXL 1.0; ML-B: FLUX.1-schnell

Embedding Methods

Spatial: LSB substitution

Replace the least-significant bit of each pixel, $k = 1$, row-major. Lossless PNG storage required.

cover pixel



bit=1

⇒

stego pixel

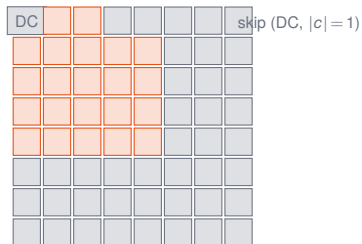


One pixel – one payload bit. Detected by RS, χ^2 , Sample Pairs.

Frequency: DCT-LSB (JSteg-style)

Flip the LSB of *non-zero* quantised AC coefficients in each 8×8 block. Skip DC and $|c| = 1$ (Westfield-pair stable). JPEG Q=95.

used (non-zero AC)

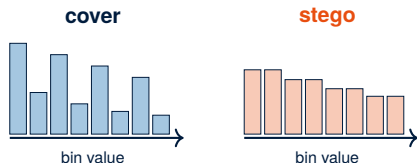


Detected by χ^2 -DCT and Calibration χ^2 .

Detector Mechanisms

Pairs-of-Values intuition

χ^2 tests the prediction that LSB embedding equalises the counts of value pairs $\{2k, 2k+1\}$. Bars within each pair converge in stego images.



χ^2 score rises whenever paired bars are mismatched in the cover and equalise in the stego.

Why training-free: no learned parameters $\Rightarrow$ AUC differences reflect carrier statistics, not training-set mismatch.

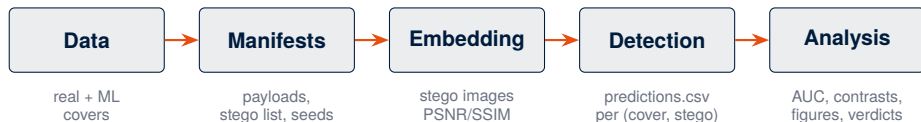
Spatial

- **RS** (Fridrich 2001): regular/singular pixel-group ratios
- χ^2 (Westfeld 1999): PoV (left)
- **Sample Pairs** (Dumitrescu 2003): traces on adjacent-pixel multisets

Frequency

- χ^2 -**DCT** (Westfeld 1999): PoV on AC coefficients
- **Calibration** χ^2 (Fridrich 2003): non-block-aligned crop + recompress reference

Pipeline Structure



Five sequential stages

- Data and manifests are reproducible from seeds.
- Embedding writes one stego file per condition.
- Detection scores cover+stego pairs.
- Aggregation derives all RQ outputs.

Run isolation

- Each run lives in `runs/<id>/`.
- Config frozen in `config.json`.
- Re-runs are idempotent: done work is skipped.

Web Platform and Live Monitoring

Architecture

- Local Python HTTP server (stdlib only)
- Single-page browser client, no build step
- Endpoints: list / launch / preview / kill / detail
- Launch spawns the runner as a subprocess

Live observability

- Subprocess stdout via Server-Sent Events
- Browser terminal panel shows progress live
- Preview validates config before any compute
- Run-detail surfaces metrics, figures, gallery

Power estimate in the drawer: Hanley–McNeil computed live in JavaScript, colour-coded against the proposal threshold $\delta = 0.05$.

Statistical Analysis

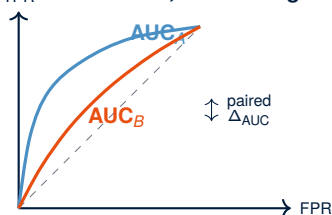
Tests

- DeLong paired ROC test for correlated AUCs (DeLong 1988)
- Bonferroni–Holm correction (confirmatory); 95% CIs (exploratory)
- Equivalence margin ± 0.025 AUC for RQ5

Power (Hanley–McNeil)

- $\alpha = 0.05$, target power 0.80; family size 15 strata per RQ
- Min detectable δ_{AUC} at $N = 1,000$ well below the $\delta_{\text{min}} = 0.05$ proposal threshold; $N = 3,000$ tightens every CI by $\sqrt{3} \approx 1.73\times$

DeLong: same cover set, two scorings



Each Δ_{AUC} comes from a single stratum; Holm controls the family-wise error rate across 15 strata.

Run Artefacts and Reproducibility

What the 3,000-group run produced

Artefact	Count
Cover images (real + ML-A + ML-B)	18,000
Stego images (3 sources $\times$ 12 conds)	108,000
Detector predictions (6 detectors)	648,000
Aggregated metrics CSVs	19
Figures (ROC panels, contrasts, forest)	61
RQ verdicts (JSON + Markdown)	2
On-disk artefacts	~ 13.5 GB
Wall-clock (cover prep + embedding + analysis)	~ 13 h

Reproducibility guarantees

- **Seeded** every stage: split, payload, embedding, ML generation. Re-running yields bit-identical covers / payloads / stego (within engine determinism).
- **Frozen config**: each run writes `config.json` with every parameter – active detectors, payload fill rates, seeds.
- **Manifest-driven**: every condition recorded in CSV manifests before execution.
- **Idempotent**: re-runs skip work already on disk; partial failures resume cleanly.
- **245 automated tests** cover detectors, embedders, RQ verdict logic, analysis pipeline.
- **Open source**: all code + figures derivable from `predictions.csv`.

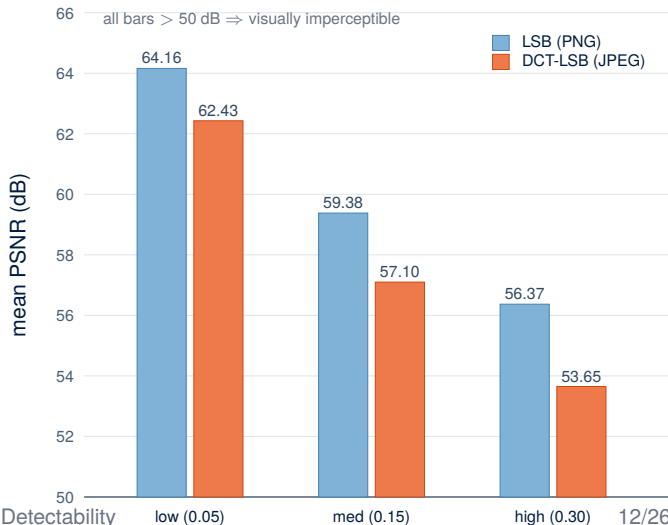
Stego Image Quality (PSNR / SSIM)

Why we track this

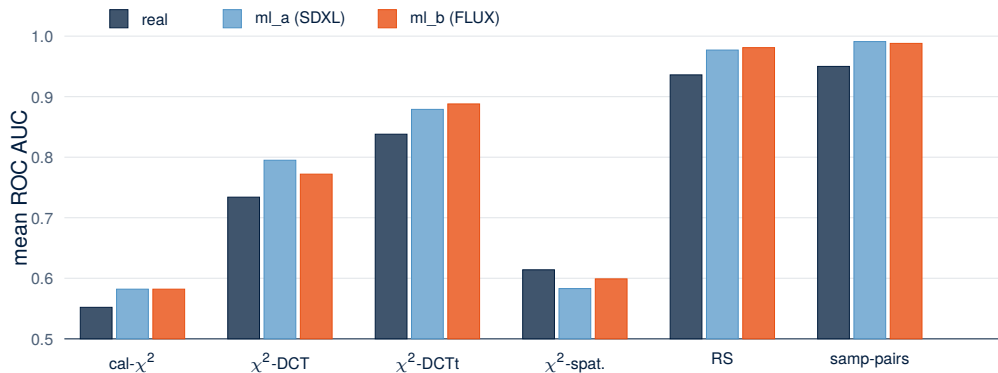
- PSNR and SSIM (Wang 2004) measure how much the stego image deviates from its cover.
- PSNR $\gtrsim 40$ dB and SSIM $\gtrsim 0.99$ are the standard “visually imperceptible” floor.
- Confirms the carrier-source effect we report is *not* a perceptual-distortion artefact.

Per-source equivalence

SSIM ≥ 0.999 everywhere. PSNR is similar across real (58.2), ml_a (59.2), ml_b (59.2 dB) – detectability differences happen at the same distortion budget.



Headline – AUC by Carrier Source and Detector



Carrier source matters, direction depends on the detector

Five of six detectors find ML carriers *easier* to attack (orange/blue > dark for $\text{cal-}\chi^2$, $\chi^2\text{-DCT}$, $\chi^2\text{-DCTt}$, RS, sample-pairs). $\chi^2\text{-spatial}$ is the lone hold-out – on LSB the diffusion outputs are *harder* to detect, a sign flip the supplementary pAUC view dissolves.

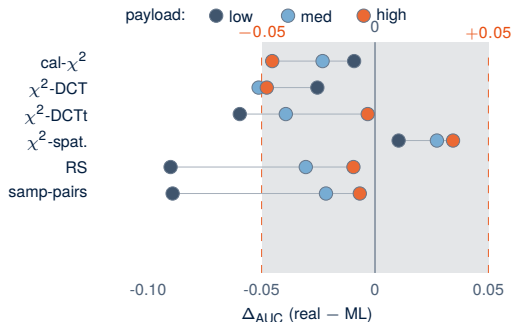
RQ1 – Real vs. Pooled ML

Test

Paired DeLong per stratum (detector $\times$ method $\times$ payload), Holm-adjusted across 18 strata (6 detectors $\times$ 3 payload levels).

Headline finding ($N=3,000$)

- Pooled Δ_{AUC} : **-0.0129**
(95% CI [-0.0137, -0.0121])
- Significant after Holm: **16 / 18** (+2, -14)
- Practically relevant ($|\Delta| \geq 0.05$): **4 / 16**
(all ML > real; 12 trivial)
- Verdict: **mixed** – direction-consistent at the practical-significance gate, all ML > real



Each detector row spans its 3 payload strata. Grey band = trivial zone $|\Delta| < 0.05$. **Four** dots clear the practical-significance threshold, all on the ML-easier side; χ^2 -spatial sits opposite.

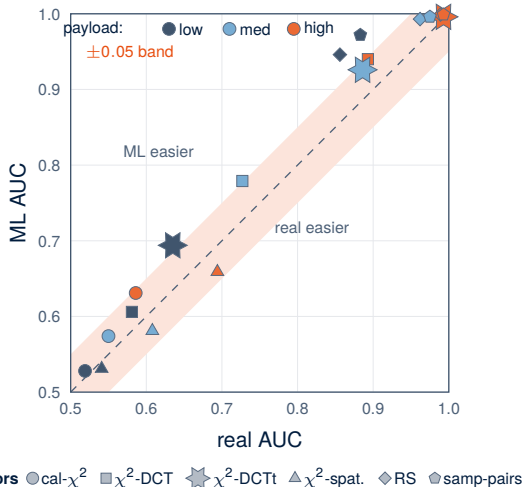
RQ1 – Per-Stratum AUC (real vs. pooled ML)

Reading the plot

- Each dot = one stratum (6 detectors $\times$ 3 payload levels = 18).
- Shape** = detector; **fill colour** = payload.
- Above the diagonal $\Rightarrow$ ML easier to detect; below $\Rightarrow$ real easier.
- Orange band = ± 0.05 practical margin.

What to look for

- All three *triangles* (χ^2 -spat.) lie below the diagonal – the lone real-easier outlier.
- The four dots outside the orange band on the ML-easier side: RS low, samp-pairs low, χ^2 -DCTt low, χ^2 -DCT med – four independent detector families agreeing.



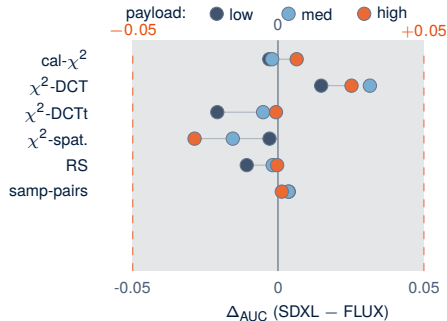
RQ2 – SDXL vs. FLUX within ML

Test

Paired DeLong on caption-matched SDXL/FLUX pairs, Holm-adjusted across 18 strata.

Headline finding ($N=3,000$)

- Pooled Δ_{AUC} : **+0.0004**
(95% CI [+0.0000, +0.0008])
- Significant after Holm: **7 / 18** (+4, -3)
- Practically relevant ($|\Delta| \geq 0.05$): **0 / 7**
- Verdict: **trivial** – the $3\times$ sample size exposes more tiny SDXL-vs-FLUX differences but none clear $\delta_{\min} = 0.05$



Every dot sits inside the trivial band – no SDXL > FLUX effect clears the $\delta_{\min}=0.05$ threshold. The generators are interchangeable in their detector signature.

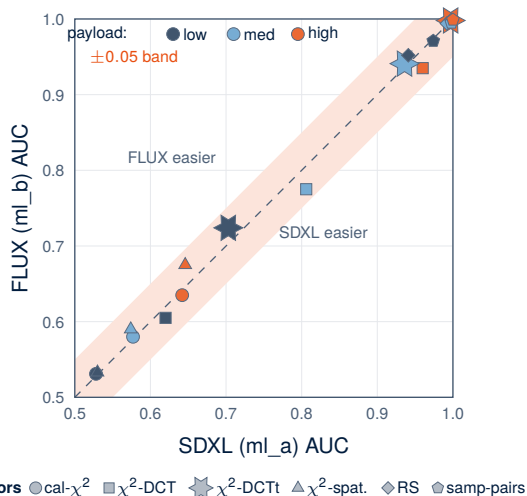
RQ2 – Per-Stratum AUC (SDXL vs. FLUX)

Reading the plot

- Each dot = one stratum (6 detectors $\times$ 3 payload levels = 18).
- **Shape** = detector; **fill colour** = payload.
- Dot on the diagonal $\Rightarrow$ generators equivalent for that stratum.
- Orange band = ± 0.05 practical margin.

What to look for

Every dot hugs the diagonal. *Squares* (χ^2 -DCT) drift slightly toward SDXL-easier, *stars* (χ^2 -DCT-tiled) and *triangles* (χ^2 -spat.) the other way; max gap ≈ 0.03 , all inside the practical band.



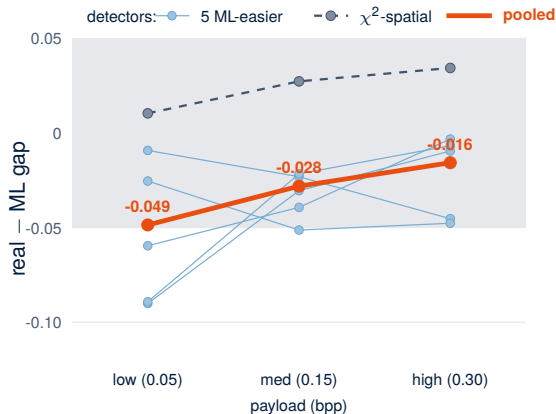
RQ3 – Payload Interaction with Source

Real – ML AUC gap, by payload ($N=3,000$)

- Low (0.05 bpp): **−0.0486**
- Medium (0.15 bpp): **−0.0281**
- High (0.30 bpp): **−0.0157**
- Trend: **monotone, magnitude decreasing with payload**
- Verdict: **supported**

Why the gap shrinks at high payload

At higher embedding rates the per-pixel modification noise dominates the carrier-source statistical fingerprint. The carrier-origin signal is strongest where the embedding signal is weakest – exactly the operationally relevant regime.



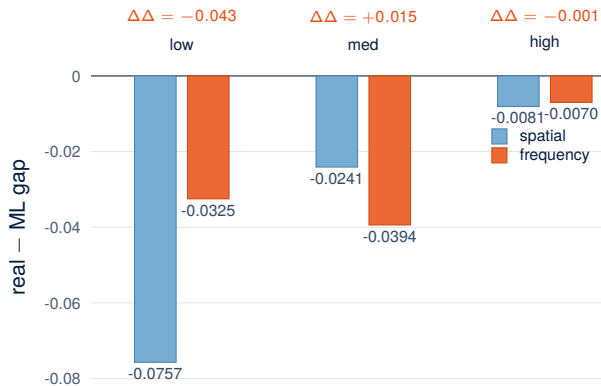
RQ4 – Spatial vs. Frequency Branch

(spatial gap – frequency gap,
 $N=3,000$)

- Pooled $\Delta\Delta_{AUC}$: **-0.0013**
(95% CI [-0.0031, +0.0005])
- Strata with 95% CI excluding 0: **2 / 3** (+1, -1)
- Practically relevant ($|\Delta\Delta| \geq 0.05$): 0 / 2
- Verdict: **trivial** – per-payload pattern persists
(spatial wins at low, frequency at medium, tie at high) but pooled $\Delta\Delta$ is now indistinguishable from zero

Per-payload pattern, pooled is trivial

Spatial dominates at low (-0.043, practical), frequency at med (+0.015, trivial), tie at high (-0.001). The 3× sample tightens error bars; adding χ^2 -DCT-tiled to the frequency-branch average closes the pooled gap.



$\Delta\Delta$ = spatial gap – frequency gap.

RQ5 – Encryption Invariance (verification)

Test

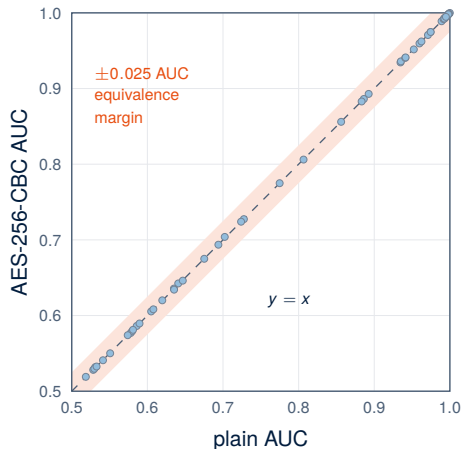
Paired DeLong between plain and AES-256-CBC stego on shared cover groups. Equivalence margin ± 0.025 AUC per stratum.

Headline finding ($N=3,000$)

- Strata within ± 0.025 : **54 / 54**
- Pooled Δ_{AUC} : -5.7×10^{-7} (95% CI ≈ 0)
- Verdict: **supported – encryption is invariant**

Why the null matters

A non-null result would indicate the detectors are reacting to payload structure rather than the embedding distortion itself. The clean null confirms the detectors respond to embedding artefacts, not plaintext patterns.



Supplementary – Partial AUC at $\text{FPR} \leq 0.10$

Why pAUC, why $\text{FPR} \leq 0.10$?

Full AUC integrates the entire ROC, including FPR regions a real steganalyst would never deploy. **pAUC** captures only the operationally relevant left tail. $\text{FPR} \leq 0.10$: standard since McClish (1989), matches the steganalysis screening regime, and fits the shape of our ROC curves.

What changes under pAUC ($N=3,000$)

- Effects **4–20 $\times$ larger**: sample-pairs/low jumps from -0.09 (full) to **-0.30** (pAUC).
- **RQ1: mixed $\rightarrow$ supported**; RQ2: trivial $\rightarrow$ mixed.
- χ^2 -spatial flip *dissolves* (all 3 strata $\leq |0.007|$, NS).

Verdict comparison (pre-reg. $\delta_{\min}=0.05$)

	full AUC	pAUC@FPR ≤ 0.10
RQ1 (real vs ML)	mixed	supported
pooled Δ	-0.0129	-0.0174
sig. / practical	16/18 ; 4/16	12/18 ; 6/12
RQ2 (SDXL vs FLUX)	trivial	mixed
pooled Δ	$+0.0004$	$+0.0008$
sig. / practical	7/18 ; 0/7	7/18 ; 1/7

Top RQ1 $|\Delta\text{-pAUC}|$ strata

sample-pairs/lsb/low: **-0.30** · rs/lsb/low: -0.16 ·
sample-pairs/lsb/med: -0.11 · rs/lsb/med: -0.09 ·
 χ^2 -DCT-tiled/dct/med: -0.06 · χ^2 -DCT/dct/high: -0.06

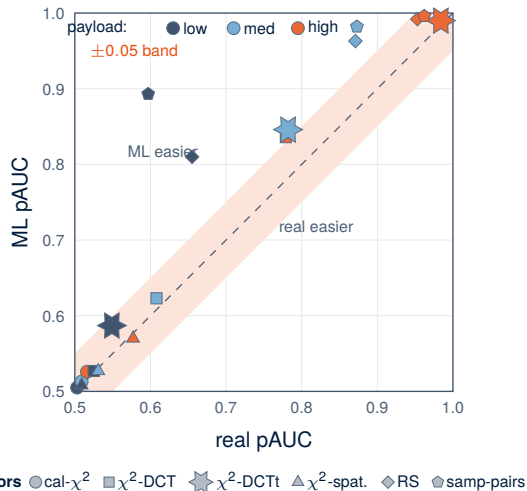
Supplementary – RQ1 Per-Stratum pAUC@FPR $\leq$ 0.10

Reading the plot

- Each dot = one stratum (6 detectors $\times$ 3 payload levels = 18).
- **Shape** = detector; **fill colour** = payload.
- Above the diagonal $\Rightarrow$ ML easier at low FPR.
- Orange band = ± 0.05 practical margin.

Same dots, sharper picture ($N=3,000$)

The points spread *much further* from the diagonal than under full AUC: sample-pairs/low and rs/low sit deep in the ML-easier zone (pAUC gap up to 0.30). The 3 triangles (χ^2 -spat.) collapse back onto the diagonal – the diffusion-flatness effect was a high-FPR artifact.



Limitations

Acknowledged deviations

- ml_b: PixArt- α replaced with FLUX.1-schnell (PixArt unavailable on HF Inference API).
- Payload fill rates revised to $\{0.05, 0.15, 0.30\}$ to avoid detector saturation at the proposal's original $\{0.25, 0.50, 0.75\}$; the lower regime is operationally more realistic and lets the carrier-source signal emerge across the full detector family.
- Verdicts shown are from the $N=3,000$ **confirmatory run**; the $N=1,000$ pre-registered run reproduces every direction with the single tightening of RQ4

Scope-bounded

- Two generator families only – cannot disentangle architecture from training data in RQ2.
- Grayscale only; no colour information.
- Classical detectors only; no learned baselines.
- RQ4 pools over an asymmetric set (3 spatial vs 2 frequency detectors); the branch contrast accounts for this in its inverse-variance weighting.

Contribution and Minimum Deliverable

What this study contributes

- Caption-matched dataset across three carrier sources under identical embedding.
- Training-free AUC-based comparison with DeLong and Holm correction.
- Pre-computed RQ verdicts and power analysis as reproducible artefacts.

Minimum deliverable met

- One encryption scheme (AES-256-CBC). ✓
- Two embedding approaches: spatial LSB and frequency DCT-LSB. ✓
- Defensible answers to RQ1 and RQ2. ✓

Phase 3 – Extensions

Completed since midway

- $N=3,000$ **confirmatory run** – $3\times$ larger sample with 6 detectors and the same factorial; verdicts above reflect this run.

Concrete next steps (~ 1 day each)

- **Lower payload tier** (0.01 / 0.02 bpp) added to the factorial via `rerun_with_existing_covers.py`; should widen the source gap on strong detectors.
- **pAUC@FPR ≤ 0.10 as co-primary** – promote the supplementary contrasts (slide 21) to a headline result.
- **JPEG-quality sweep** ($Q \in \{75, 85, 95\}$): does the carrier-source effect survive lossy recompression?
- **Vectorise the analysis pipeline** – the $O(N^2)$ pure-Python ROC computation took ~ 6 h on the 648,000-row prediction set;

Mechanism investigations

- **PoV-histogram skewness per source**: measure $\sum_k |n_{2k} - n_{2k+1}|$ on cover histograms of real vs. ML directly, with no embedding. Validates the diffusion-flatness explanation for the χ^2 -spatial flip.
- **KL or MMD between source distributions vs. Δ_{AUC}** : per detector, plot carrier-statistic distance against the observed effect size. Turns a per-detector heuristic story into a quantitative one.

Smaller scope cleanups

- **Colour mode**: store covers as YCbCr-Y instead of grayscale; detectors stay grayscale-equivalent but the carrier-statistic distance changes.
- Promote the χ^2 -spatial finding to a discussion paragraph in the report with the PoV-skewness evidence above.

Thank you

Questions and Discussion

Code and run artefacts: `runs/<id>/`
